Methodology guide
PVMD Toolbox

Developed by:
Photovoltaic Materials and Devices
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Starters Manual PVMD Toolbox

by

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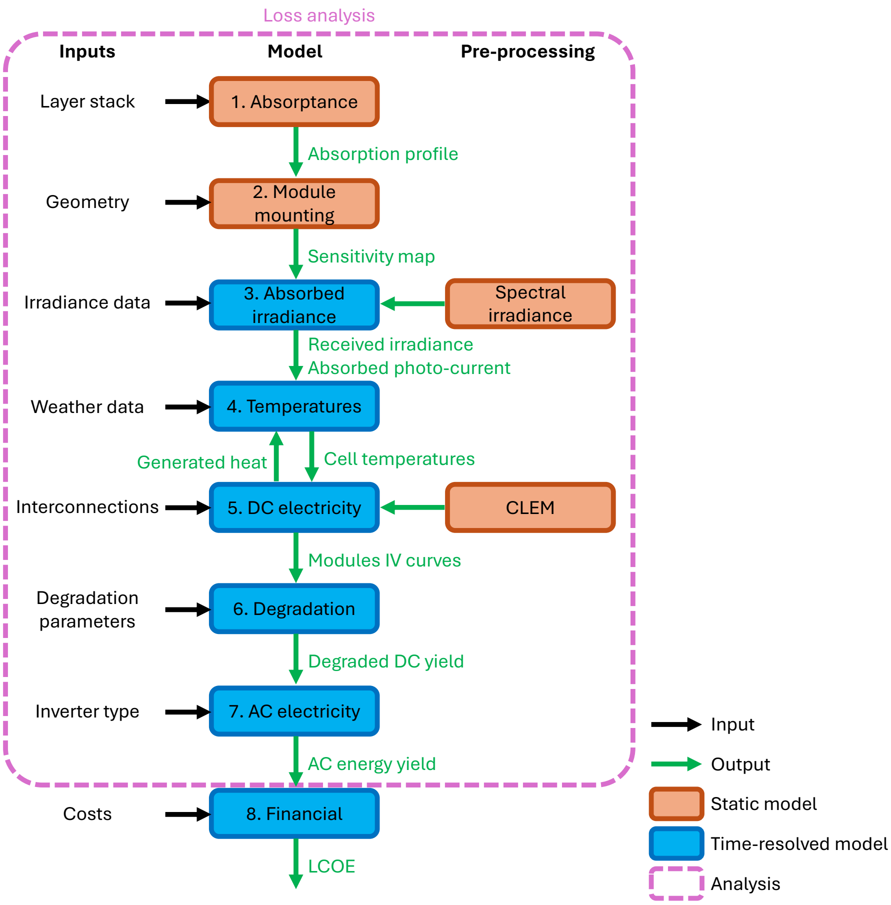


Figure 1.1: PVMD Toolbox Workflow Diagram

2. Cell

The first step in simulating the energy yield of a PV system involves calculating the absorptance profile of the absorber layers within the device. This step requires a detailed description of the layer stack, including the thickness and complex refractive index of each layer. Additionally, the presence of surface texturing at various interfaces must be specified. Another essential input is the identification of absorber layers within each sub-cell, as well as the assignment of each sub-cell to its corresponding sub-module. Figure 2.1 illustrates three example layer stacks used as input, along with their resulting absorptance profiles.

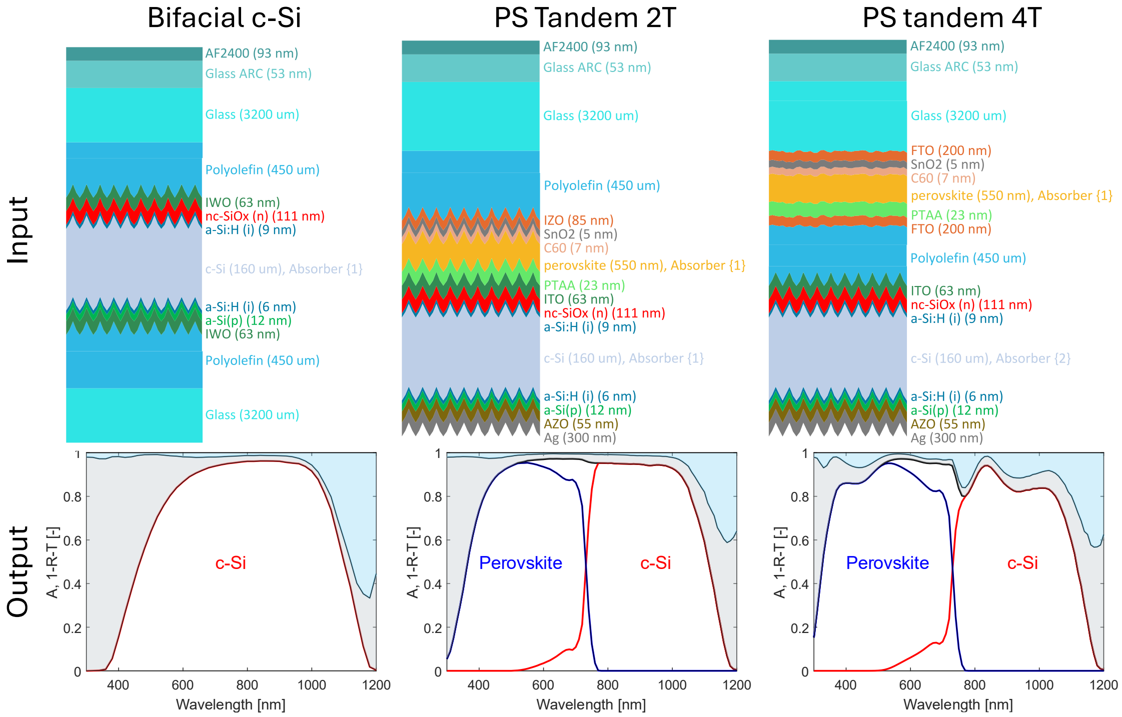


Figure 2.1: Three examples of the input and output of the first simulation step. The input contains the layer stack with the thickness and complex refractive index of all layers. Also the absorber layers of the sub-cells are indicated with "Absorber", and its corresponding submodule is indicated in curly brackets. The output contains the spectral absorptance profile of the absorber layers. The reflection and parasitic absorption from supporting layers are indicated in light blue and light grey, respectively. Our framework can handle bifacial modules too, for which also transmission losses can be spectrally quantified.

The absorptance profile is computed using GenPro4 [1], which is integrated into the PVMD Toolbox. GenPro4 combines the net radiation method [2] with ray-tracing techniques to account for surface texturing. Besides pyramid texturing that is common for wafer-based modules, various nano-textures can also be simulated, such as Asahi U-type texturing [3] that is used in the top part of the 4T module.

While Figure 2.1 presents the spectral absorptance for normally incident light, the simulation also evaluates absorptance across a range of incidence angles. These angle-resolved profiles, provided in the Supporting Information, are crucial for accurately modeling the optical response of the device under real-world illumination conditions. Additionally, bifacial modules also require an absorptance profile for light entering from the rear side. This information serves as a foundation for the subsequent simulation steps.

3. Module

The optical response calculated in the first simulation step is used to generate the so-called *sensitivity map* [4] of the PV modules. This map quantifies how sensitive each cell in the module is to light originating from different regions of the sky. To construct the sensitivity map, the sky is discretized into multiple angular segments, as illustrated in Figure 3.1a. A sensitivity value $S_{i,j,k}(\lambda)$ is generated for each sky segment i, each cell j in the module, and each sub-cell k and is wavelength dependent. $S_{i,j,k}(\lambda)$ is defined as the fraction of the incident light from the sky segment that is absorbed in the absorber layer [4], written as

$$S_{i,j,k}(\lambda) = \frac{I_{abs,i,j,k}(\lambda)}{I_{sky,i}(\lambda)}, \tag{3.1}$$

where $I_{abs,i,j,k}(\lambda)$ is the absorbed irradiance and $I_{sky,i}(\lambda)$ is the incident irradiance from sky segment i. This sensitivity is inherently wavelength-dependent due to the spectral characteristics of both the absorptance profile and the ground's albedo reflectance.

The calculation of the sensitivity map considers two levels of detail. At a fine level, nearby objects are considered that have different impacts of different cells in the system. A critical input for this level is the geometric configuration of the PV modules and their surroundings. This includes the dimensions and spectral reflectance of nearby objects, as well as the albedo of the ground. Previous work [5] has shown that ground material properties significantly influence the performance of bifacial tandem devices. Figure 3.1b presents two examples of sensitivity maps for a c-Si single-junction module, showing the fraction of light, compared to the incident light from the sky elements, absorbed in the bottom-left cell at a wavelength of 600 nm. It should be noted that these maps only represent a cross-section of the full sensitivity dataset, which spans all cells and wavelengths.

The PVMD Toolbox offers two methods for generating sensitivity maps, differing in complexity and computational cost. The most detailed approach uses the forward ray-tracing software LUX [4], which employs a Monte Carlo method and is well-suited for periodic scenarios, such as field-deployed PV modules. However, LUX can become computationally intensive for complex geometries or high ray counts.

As a faster alternative, the Toolbox includes a backward ray-tracing method developed by Calcabrini *et al.* [6]. This method combines view factor calculations with single-generation ray-tracing, offering a balance between accuracy and efficiency. A comparison of both methods is provided in the Supporting Information.

Importantly, both approaches support the simulation of bifacial modules. Sensitivity maps can be generated for both the front and rear sides, enabling accurate modeling of rear-side irradiance contributions.

Besides the effects captured by the ray-tracing models, also skyline effects can be considered at a more coarse level. This includes all objects at a distance far enough that they impact all cells in the system equally. The skyline is described by a function that expresses the height of the skyline for each azimuth, as shown in Figure 3.1c. Irradiance coming from sky elements below the skyline cannot be absorbed by the cells, so the sensitivity values for these sky elements are manually set 0. These objects are not included in the ray-tracer, as the increased amount of data needed to describe the environment will drastically increase the computational time of the ray-tracing algorithm. Given that these effects appear at a far distance, they will impact all cells equally, justifying this fast approach.

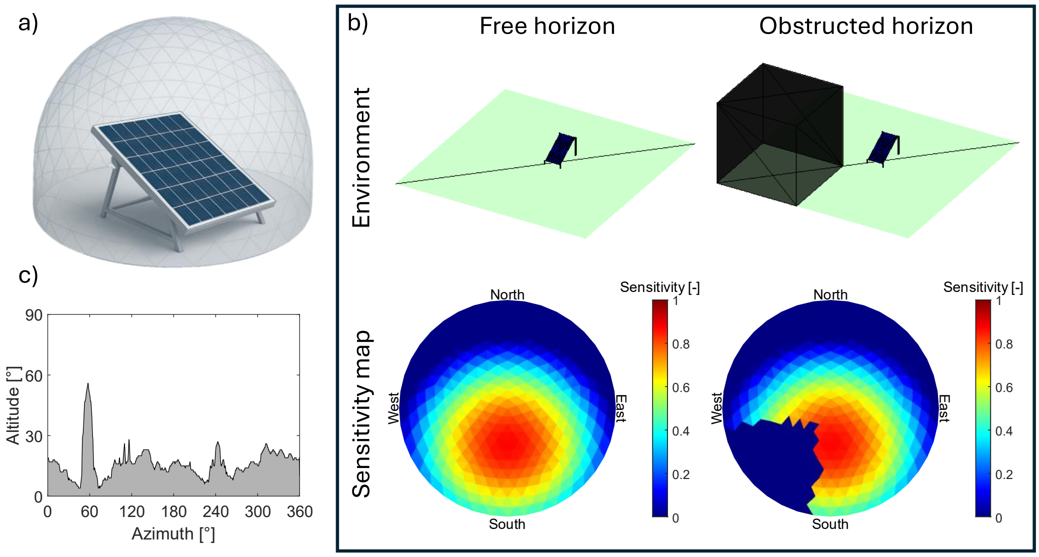


Figure 3.1: a) shows visualization of the sensitivity map. The sky is discretized into different sky segments for which its sensitivity is calculated. b) shows two examples of a sensitivity map of two scenario's. It can be seen that the object has a significant impact in the sensitivity map. The two maps are calculated for single junction c-Si modules and the cross section of the bottom left cell at a wavelength of 600 nm is shown. The sensitivity maps are generated with the backward ray-tracer by Calcabrini *et al.* [6].

4. Weather

The first time-resolved step in the PVMD Toolbox involves calculating the absorbed irradiance and the resulting photo-current density for each cell in the PV module. As illustrated in Figure 4.1, this is achieved by combining the previously computed sensitivity map with a time-dependent *sky map*.

The sky map describes the angular distribution of irradiance across the sky dome for each simulation timestep [4, 7]. It is generated using the Perez model [8], which requires as input the solar position, direct normal irradiance (DNI), and diffuse horizontal irradiance (DHI). These meteorological inputs are obtained via Meteonorm [9].

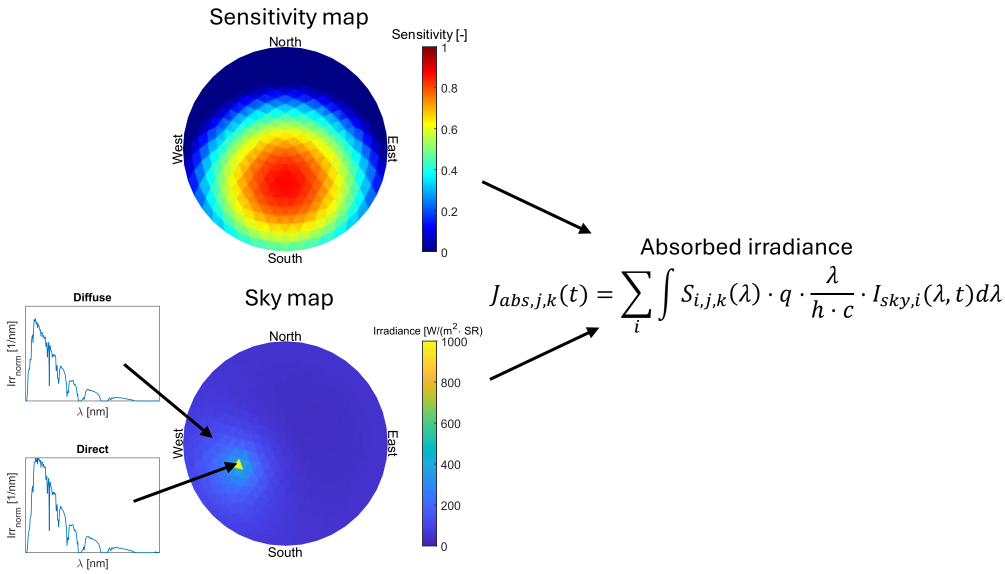


Figure 4.1: An example of the calculation of the absorbed irradiance for a PV module in Delft on the 1st of June at 16:00. The sensitivity map and the sky map are combined to calculate the absorption for each cell at each hour.

The Perez model provides the total irradiance per sky segment but does not include spectral information. As the spectral decomposition is critical for accurately modeling multi-junction devices, the PVMD Toolbox integrates two spectral models: SMARTS [10] and SBDART [11]. These models are used to generate look-up tables of normalized spectral distributions for both direct and diffuse components as a function of air mass, corresponding to the first pre-processing block in Figure 1.1. For each timestep, the irradiance values from the sky map are multiplied by the correct normalized spectrum (depending on whether the sun is within a given sky segment) to obtain the full spectral irradiance across the sky.

A key distinction between the two spectral models is that SMARTS assumes clear-sky conditions, while SBDART accounts for cloud cover. To incorporate cloud effects, SBDART generates separate look-up tables for three sky conditions: clear, partly cloudy, and overcast. The appropriate table is selected based on the sky clearness index provided by the Perez model [8]. The complete look-up tables for both models are provided in the Supporting Information.

5. Thermal

Following the calculation of absorbed irradiance, the PVMD Toolbox simulates the cell temperature throughout the year. Initially, the Toolbox employed only the Fuentes fluid dynamic model [12]. In its current version, the framework has been extended to include additional thermal models, such as the Duffie-Beckman model [13], the Sandia Module Temperature model [14], and the Faiman model [15]. This allows users to select the most appropriate model for their specific application. Notably, the Fuentes model remains particularly suitable for emerging PV systems, as it does not rely on empirical parameters [7]. A comparative analysis of these models is provided in the Supporting Information.

A limitation of earlier implementations was the assumption of steady-state conditions for cell temperature. However, due to the thermal mass of PV modules, temperature changes occur gradually and must be modeled accordingly, especially at high temporal resolutions [16]. To account for this, the cell temperature at time t, denoted $T_{cell}(t)$, is calculated using

$$T_{cell}(t) = T_{cell,ss} + e^{-\frac{t_{int}}{\tau}} \left(T_{cell}(t - t_{int}) - T_{cell,ss} \right), \tag{5.1}$$

where $T_{cell,ss}$ is the steady-state temperature from the thermal model, t_{int} is the simulation time step, and τ is the thermal response time, assumed to be 7 minutes [16]. This equation takes a weighted average between the steady-state temperature under current conditions and the temperature of the previous time step. The weight given to the previous time step decreases exponentially with the duration of the timestep.

Another enhancement addresses the effect of reverse bias on cell temperature. In the fluid dynamic model, one of the heat flow terms is the electrical power generated by the cell, P_{gen}, which is subtracted from the total absorbed power as it does not contribute to heating. Previously, P_{gen} was estimated to be:

$$P_{gen} = I_{abs} \cdot \eta_{STC}, \tag{5.2}$$

where I_{abs} is the absorbed irradiance and η_{STC} is the efficiency under standard test conditions (STC). However, cells almost never operate at STC condition during outdoor operation. Furthermore due mismatches in irradiances within the module and current matching requirements, cells can operate at non-ideal conditions and even reach reverse biases. The latter would mean that a cell is dissipating heat instead of removing heat. These aspects make Eq. (5.2) not always accurate.

To address this, an iterative approach is implemented. The electrical operating point of each cell is determined in the electrical simulation step, and the thermal and electrical calculations are repeated until both T_{cell} and P_{gen} converge [17], as illustrated in Figure 1.1. In the first iteration, P_{gen} is estimated using Eq. (5.2), and in subsequent iterations, it is updated using the output from the electrical simulation. The Supporting Information shows the maximum temperature and module power at different iterations for a test case with partial shading. Typically, convergence is achieved within three iterations.

6. Electrical

The electrical simulation in the PVMD Toolbox is structured in two levels. First, the current-voltage (IV) characteristics of all cells or sub-cells within the module are modeled individually. These cell-level IV curves are then combined to construct the overall IV curve of the complete PV module.

Sub-cell IV curves

The electrical behavior of each sub-cell is modeled separately for the forward bias ($V_{cell} > 0$) and reverse bias ($V_{cell} < 0$) regimes. In forward bias, the one-diode equivalent circuit model is employed. This model consists of a current source, a diode, and two resistive elements. The cell current I_{cell} as a function of voltage V_{cell} is implicitly defined by

$$I_{cell}(V_{cell}) = I_{ph} - I_0 \cdot \left(e^{\frac{V_{cell}+I_{cell}\cdot R_s}{n\cdot V_t}} - 1\right) - \frac{V_{cell} + I_{cell} \cdot R_s}{R_{sh}}, \tag{6.1}$$

where I_{ph} is the photo-generated current, I_0 and n are the diode's saturation current and ideality factor, and R_{sh} and R_s are the shunt and series resistances, respectively.

To determine these parameters, the PVMD Toolbox uses the *Calibrated Lumped Element Method* (CLEM) as a second pre-processing step to significantly reduce computational complexity. [7]. This CLEM model requires as input either measured or simulated IV curves of the (sub-)cell under varying temperatures and irradiance levels, such that it can derive the dependencies of the parameter value on the cell temperature and absorbed photo-current. These dependencies are used to obtain a function for each parameter which depends on cell temperature and absorbed current. A more detailed explanation on how the CLEM model makes these functions is provided in the Supporting Information. It should be noted that the level of detail considered in the electrical simulation, and whether certain effects or transport mechanisms are included, depend on how the input IV curves are obtained. These input IV curves can be obtained in various ways, ranging from detailed semiconductor simulations in TCAD Sentaurus [18] to measured solar cells.

The PVMD Toolbox uses the pre-processed functions from the CLEM model and inputs the T_{cell} and J_{abs} from step 3 and 4, respectively, to reconstruct the IV curves of the sub-cells at different operating conditions. In case IV curves at varying conditions are unavailable, the CLEM model also allows for datasheet values as input. The details for the translation of datasheet values into parameter dependencies are also provided in the Supporting Information.

At reverse bias conditions, we utilize the model from Alonso-Garcia *et al* [19], which expresses I_{cell} as

$$I_{cell}(V_{cell}) = \frac{I_{ph} - R_{sh} \cdot V_{cell} + c \cdot V_{cell}^2}{1 - \exp\left(B_e\left(1 - \sqrt{\frac{\phi_T - V_b}{\phi_T - V_{cell}}}\right)\right)}, \tag{6.2}$$

where c is a coefficient to characterize the parabolic behavior of the current, B_e is a quasi-constant parameter, ϕ_T is the built-in junction voltage, and V_b is the breakdown voltage.

In multi-junction devices, it is possible for electron-hole pairs that recombine radiatively in a high bandgap sub-cell to be absorbed in lower-bandgap sub-cells, known as photon recycling [20–23]. This is implemented using the method by Jäger *et al.* [24], as previously described in [25]. The IV curve of lower-bandgap sub-cell is adjusted by updating the photo-generated current of sub-cell k ($I_{ph,k}$) with

$$I_{ph,k} = I_{ph0,k} + \sum_{n=1}^{k-1} \eta_{LC,n\to k} \cdot (I_{ph,n} - I_{cell,n}(V_{cell,n})), \tag{6.3}$$

where $I_{ph0,k}$ is the initial photo-generated current of sub-cell k as outputted by the CLEM, $\eta_{LC,n\rightarrow k}$ is the luminescence efficiency from sub-cell n to sub-cell k [25], and $I_{cell,n}(V_{cell,n})$ is the current of sub-cell n at its operating voltage.

Finally, the model accounts for *meta-stability* effects, which are particularly relevant for perovskite sub-cells. These effects manifest as reversible efficiency changes during light and dark cycles [26, 27]. The implementation follows the model by Remec *et al.* [27], which introduces an initial voltage loss V_{loss} that recovers under illumination. V_{loss} is calculated with

$$V_{loss} = A_{MS} \cdot e^{-\frac{H_{cum}}{\tau}}; \frac{1}{\tau} = B_{MS} \cdot e^{-\frac{E_{a,MS}}{k_b \cdot T_{cell}}}, \tag{6.4}$$

where A_{MS}, B_{MS}, and $E_{a,MS}$ are calibration constants, H_{cum} is the cumulative daily irradiance, and k_b is the Boltzmann constant.

Module IV curve

Once the IV characteristics of all individual cells and sub-cells are determined, they are combined to construct the overall IV curve of the PV module. The specific method for combining these curves depends on the module's interconnection scheme. Fundamentally, this involves summing voltages for series connections and summing currents for parallel connections.

For a series connection, the total voltage as a function of current can be written as

$$V_{total}(I) = \sum_i V_i(I), \tag{6.5}$$

where $V_i(I)$ represents the voltage of segment i as a function of current.

Conversely, for a parallel connection, the total current as a function of voltage is expressed as

$$I_{total}(V) = \sum_i I_i(V), \tag{6.6}$$

where $I_i(V)$ is the current of segment i as a function of voltage.

Figure 6.1 illustrates several interconnection schemes supported by the PVMD Toolbox, highlighting its flexibility. For single-junction devices, various layouts are possible (Figure 6.1a), each offering different levels of shade tolerance. The standard configuration consists of a series connection of cells with bypass diodes. The effect of these diodes is modeled by modifying Eq. (6.5) as

$$V_{total} = \max\left(\sum_i V_i(I), -V_{diode}\right), \tag{6.7}$$

where V_{diode} is the forward voltage of the bypass diode.

To enhance shade tolerance, a butterfly layout can be used. This design employs half-cut cells arranged into two substrings connected in parallel with one bypass diode. For even greater resilience, bypass diodes can be replaced with buck converters, forming so-called *smart modules*, as proposed by Mirbagheri Golroodbari *et al.* [28]. These bypass diodes allow each substring to work at its own maximum power point, allowing for a greater electricity production in shaded conditions. The implementation of these converters is detailed in the Supporting Information.

Multi-junction devices can come in different terminal configurations [29–32], significantly influencing the module interconnection scheme. In previous work [33], we compared the performance of two-terminal, three-terminal, and four-terminal PV modules under outdoor conditions. The PVMD Toolbox also supports simulations of triple-junction devices, as demonstrated in [25].

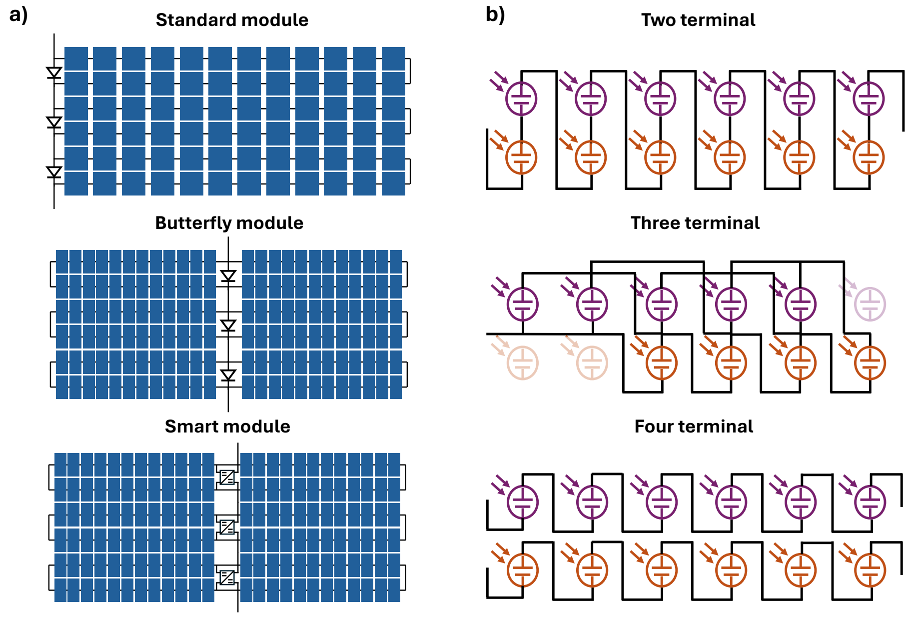


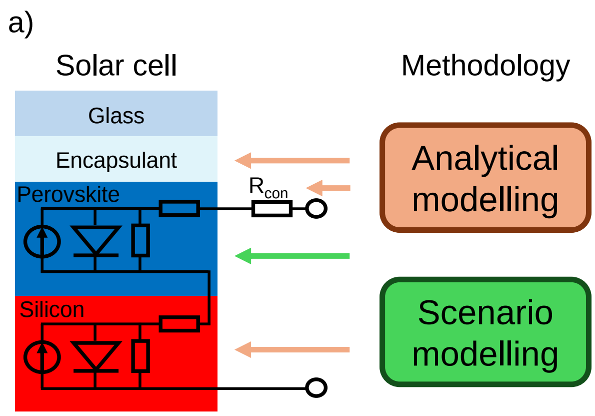
Figure 6.1: Some examples of the different module layouts that can be simulated in the PVMD Toolbox. a) shows different interconnection schemes for single-junction cells, and b) illustrates the different topologies for multi-junction devices [25].

7. Degradation

To identify tolerable values for k_{per}, we extend the PVMD Toolbox [7] by developing a methodology to simulate degradation. The PVMD Toolbox is a modeling framework designed to predict the EY of various PV systems by simulating key aspects of their performance. The simulation process begins by calculating the optical response of each subcell for the relevant wavelength range (300-1200 nm), which helps determine the incoming irradiance. This irradiance data is then used to calculate the temperature of each cell throughout the year. Finally, the electrical performance is simulated to obtain the EY of the PV module and the system afterwards. Full details and validation of the PVMD Toolbox have been documented in previous studies [7, 25, 34, 35].

To predict the lifetime energy yield (LEY) of a PV system, we extend the annual PVMD simulation by adjusting input parameters to account for degradation. Figure 7.1a illustrates the degradation modeling methodology.

We use an analytical model to simulate degradation in the silicon bottom cell, encapsulation, and contact resistance. Since these components are similar to those in single-junction c-Si modules, available outdoor data can be used to calibrate the model effectively.



b)

Mechanism	Stress factor	Time profile	Effect on module
Discoloration	UV, T	Constant	Encapsulant
MID	RMC, T	Increase	Bottom cell
TC	T	Decrease	R_{con}
LID	J_{ph}	Decrease	Bottom cell

Figure 7.1: a) An overview of the methodology, where a scenario-based approach is used for degradation in the top cell, and an analytical approach is used for the bottom cell, encapsulant, and contact resistance. b) An overview of the degradation mechanisms considered for the analytical model, including their stress factors, time profile, and effect on the module. MID, TC, LID, and RMC stand for moisture-induced degradation, thermal cycling, light-induced degradation, and relative moisture content.

For the perovskite top cell, a different approach is required. Due to the lack of large-scale outdoor data for perovskite modules, we adopt a scenario-based approach. The simulation of the top cell is adjusted based on three different scenarios to explore tolerable degradation rates.

The performance of the top and bottom cell is modeled using a one-diode equivalent circuit model with two resistors [36–39]. The behaviour of this model can be described with 5 parameters, namely the photo-generated current (I_{ph}), the diode saturation current (I_0), the diode ideality factor (n), the shunt resistance (R_{shunt}), and the series resistance (R_s). To simulate a tandem cell, both circuits are connected in series with an additional resistor to represent the interconnection (R_{con}).

7.0.1 Analytical modeling

In literature, various analytical models have been developed to predict performance loss in c-Si modules [40, 41]. However, these models typically produce only a single value for k without distinguishing between losses in voltage or current. For this reason, they are unsuitable for tandem devices.

To address this limitation, we develop a new analytical model that not only provides k but also simulates losses in the absorption profile and the current-voltage (IV) curve. This model accounts for different degradation types and stress factors, simulating their impact on module performance through analytical equations. The degradation mechanisms included in this model are discoloration, moisture-induced degradation (MID), thermal cycling-induced degradation (TC), and light-induced degradation (LID), as these mechanisms are most severe or occurring most frequently [42, 43].

Certain effects, such as hotspots, potential-induced degradation, and soiling, are excluded from this study as they are not compatible with the modeling approach. Hotspots are due to current inhomogeneity within the cells, which can result from partial shading or wafer impurities [44]. However, since the PVMD Toolbox assumes all cells are identical and, in this study, considers an unobstructed horizon, hotspot effects cannot be incorporated into the model. Similarly, PID occurs due to a potential difference between the module frame and the active circuit, which can reach up to 1500 V when multiple modules are connected in series [45]. This phenomenon cannot be captured in the simulations, as the model considers only the direct current (DC) performance of a single module, where such high potential differences are not present. Finally, the effect of soiling [46] has also been excluded from this model, as rainfall is not included as input parameter of the PVMD Toolbox, and cleaning strategies would need to be considered, making it a complex degradation mechanisms to model. It would be interesting for future work, however, to study how soiling impacts the current matching in tandem devices, as soiling effects are not the same across all wavelengths [47]. Nevertheless, the exclusion of these mechanisms does not lead to an underestimation of degradation, as the final calibration is based on real-world outdoor PV system data. Moreover, similar analytical models in the literature [40, 41] also do not account for these more complex degradation mechanisms.

Each degradation mechanism is characterized by its stress factors, time profile, and impact on the module. Figure 7.1b provides an overview of the degradation mechanisms considered in this study. The stress factors considered in the model are ultraviolet (UV) irradiance, the temperature of the module (T), the relative moisture content (RMC), and the photo-generated current (J_{ph}). Another important characteristic is the time profile of a degradation mode. Degradation rates are not necessarily constant over time; they may increase or decrease throughout the module's lifespan [48]. Finally, the degradation modes are characterized by which part of the module they affect.

For each mechanism, a separate degradation rate k_i is calculated. To obtain the power loss ($P_{loss,i}$) solely due to a mechanism i, we use

$$P_{loss,i}(t) = \int_0^t k_i(\tau)d\tau. \tag{7.1}$$

It should be realized that all mechanisms affect the performance differently at the same time. This means that the total P_{loss} does not necessarily equal the sum of the individual power losses.

Discoloration

The encapsulant in PV modules, typically made of ethylene vinyl acetate (EVA) or polyolefin (POE) [49], undergoes color changes over its lifetime. Exposure to UV radiation can alter the encapsulant's chemical composition, impacting the optical performance of the device [50].

To predict the performance loss due to discoloration, we use the modified Arrhenius equation from Sinha *et al.* [51], which calculates the discoloration degradation rate (k_{dis}) as:

$$k_{dis} = A_{dis} \cdot e^{\frac{-E_{a,dis}}{k \cdot T}} \cdot (\mathrm{UV})^{n_{dis}}, \tag{7.2}$$

where A_{dis}, $E_{a,dis}$, and n_{dis} are the pre-exponential factor, the activation energy, and the exponent related to the discoloration, respectively, and k is the Boltzmann constant. Based on the findings in literature, we assume n_{dis} to be 1 [52] and $E_{a,dis}$ to be 0.39 eV [51]. The value of A_{dis} is determined through calibration later in this section. The amount of UV is defined as all in-plane irradiance with a wavelength smaller than 400 nm [53]. Although the UV and T have seasonal fluctuations, they are approximately similar for all years. Therefore k_{dis} is constant throughout the lifetime of the PV module, being consistent with literature [48].

Since discoloration affects the encapsulant's optical properties, we model performance loss by adjusting the refractive index ($N(\lambda) = n(\lambda) + i \cdot k(\lambda)$) of the encapsulant in the EY simulations. The PVMD Toolbox uses $N(\lambda)$ for each material to calculate the device's optical response [7].

To model changes in the refractive index due to discoloration, we use data from Meena *et al.* [54], who measured variations in internal quantum efficiency and reflection spectra of discolored modules. Figure 7.2 illustrates different $N(\lambda)$ for different values of P_{loss} and the corresponding the optical response. This change in optical response will then also have an impact on the thermal and electrical simulation step of the PVMD Toolbox.

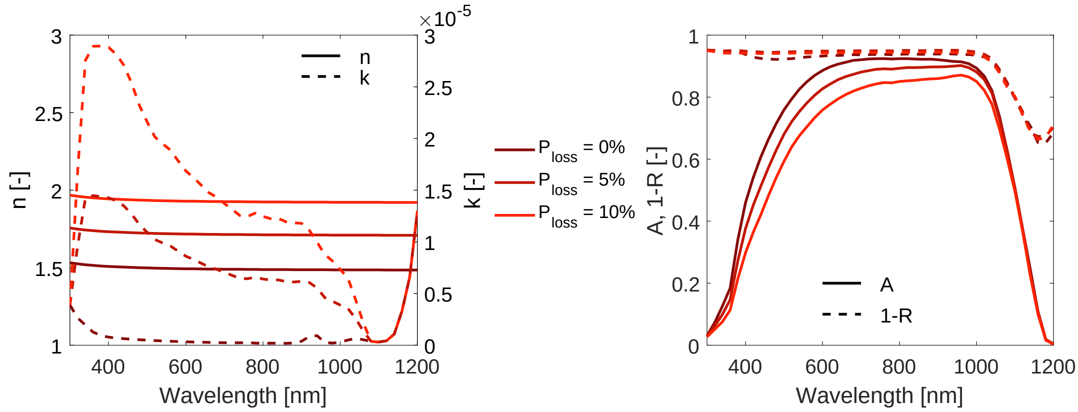


Figure 7.2: The change in $N(\lambda)$ and its effect on the absorption profile for different values of P_{loss}, solely due to discoloration.

Moisture induced degradation

Moisture ingress in PV modules can trigger various degradation modes, such as delamination and corrosion [55, 56]. While these mechanisms are distinct, it is challenging to separate them due to their same stress factor. Therefore, their effects are combined in our analysis.

To quantify the amount of degradation due to MID (k_{MID}), the Peck model is used [40, 41, 57, 58]:

$$k_{MID} = A_{MID} \cdot e^{-\frac{E_{a,MID}}{k \cdot T}} \cdot (\mathrm{RMC})^{n_{MID}}, \tag{7.3}$$

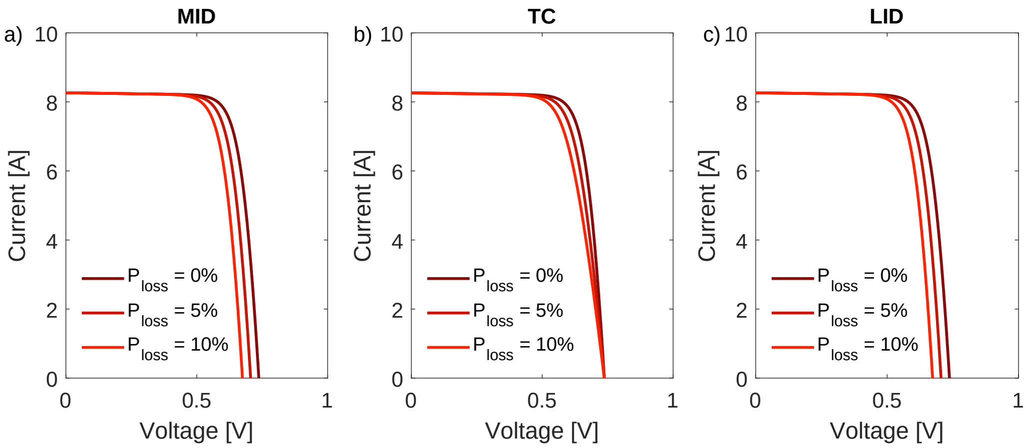


Figure 7.3: The effect on the IV curve of different values of P_{loss}, solely due to MID (a), TC (b), and LID (c).

where A_{MID}, $E_{a,MID}$, and n_{MID} are the pre-exponential factor, the activation energy, and the exponent related to MID, respecitvely. The RMC is the relative moisture content that describes the amount of moisture that has entered the module compared to the saturation concentration of the encapsulant. In previous work, a model has been developed to simulate and predict the RMC over time for different climates [58]. The values for $E_{a,MID}$ and n_{MID} are assumed to be 0.809 eV and 1.87 based on previous work [58], and A_{MID} will be used for calibration. Since moisture ingress is a gradual process, RMC increases over time. As a result, the degradation rate k_{MID} also increases throughout the module's lifetime, consistent with literature findings [48].

As MID affects both the electrical and optical properties of PV modules, its impact is simulated by adjusting the diode model parameters of the c-Si bottom cell. Zhu *et al.* [59] conducted damp heat tests to measure the effects of moisture ingress on IV curves. They also identified how the parameters of the equivalent circuit model change due to moisture exposure. These trends are used to adjust the simulation of c-Si performance, incorporating the associated performance loss. Figure 7.3a illustrates the changes in the IV curve for different values of P_{loss} caused by MID.

Thermal cycling

In addition to the absolute temperature influencing Arrhenius processes in discoloration and MID, temperature fluctuations over time introduce additional degradation mechanisms. High-temperature variations, combined with mismatches in the thermal expansion coefficients of different materials, cause layers to expand at different rates, leading to thermomechanical stress on the interconnections [60, 61]. Bosco *et al.* [62] proposed an analytical equation that can describe the damage due to TC (D_{TC}), written as

$$D_{TC} = A_{TC} \cdot (\Delta T)^{n_{TC}} \cdot R^b \cdot e^{-\frac{E_{a,TC}}{k \cdot T}}, \tag{7.4}$$

where A_{TC}, $E_{a,TC}$, and n_{TC} are the pre-exponential factor, the activation energy, and the exponent related to TC, respectively, ΔT is the average daily temperature fluctuation, R is the number of times the temperature has fluctuated, and b is an exponent to R. The values for $E_{a,TC}$, n_{TC}, and b

are assumed to be 0.12 eV, 1.9, and 0.33 as calibrated by N. Bosco *et al.* [62], and A_{TC} will be used for final calibration.

As Equation (7.4) outputs the total amount of damage instead of a degradation rate, the degradation rate due to TC (k_{TC}) can be obtained iteratively with

$$k_{TC}(t) = \frac{D_{TC}(t) - \sum_{\tau=1}^{t-1} k_{TC}(\tau) \cdot \Delta t}{\Delta t}, \tag{7.5}$$

where Δt is the time step. Given that b is equal to 0.33 and R will constantly increase over time, D_{TC} will keep rising but at a slower pace, causing k_{TC} to decrease over time. It should be noted that Equation (7.5) applies only for $t > 0$. $k_{TC}(0)$ is defined to be 0%/year.

Since TC degradation primarily affects the contacts [60–62], we adjust the value of R_{con} to simulate the required P_{loss}. Figure 7.3b illustrates the impact of TC degradation on the IV curve.

Light induced degradation

The final degradation mechanism considered is light-induced degradation (LID), where the efficiency of the solar cell decreases due to prolonged exposure to light. According to literature, this is attributed to an increase in the minority carrier lifetime (τ_{min}) within the solar cell [63–65], with the J_{ph} acting as the stress factor [66, 67].

Experimental studies have demonstrated that τ_{min} decays exponentially with exposure to J_{ph} and eventually saturates at a constant value [67, 68]. This time-dependent behavior of $\tau_{min}(t)$ can be modeled as:

$$\tau_{min}(t) = \tau_{min,\infty} + (\tau_{min,0} - \tau_{min,\infty}) \cdot e^{\frac{-\int_0^t J_{ph}(\tau) d\tau}{A_{LID}}}, \tag{7.6}$$

where $\tau_{min,0}$ and $\tau_{min,\infty}$ are the minority carrier lifetime initially and after saturation, and A_{LID} is the decay constant for LID.

Since τ_{min} and I_0 are inversely proportional [69], the value of I_0 can be adjusted using

$$I_0(t) = \frac{I_{0,init}}{C_\tau(t)}, \tag{7.7}$$

where $I_{0,init}$ is the initial value of I_0 and $C_\tau(t)$ is the ratio of $\tau_{min}(t)$ over $\tau_{min,0}$, written as

$$C_\tau(t) = \frac{\tau_{min}(t)}{\tau_{min,0}} = C_{\tau,\infty} + (1 - C_{\tau,\infty}) \cdot e^{\frac{-\int_0^t J_{ph}(\tau) d\tau}{A_{LID}}}, \tag{7.8}$$

where $C_{\tau,\infty}$ is $\frac{\tau_{min,\infty}}{\tau_{min,0}}$. Based on measurements in literature [68], $C_{\tau,\infty}$ is assumed to be 0.5, and A_{LID} is used for calibration. The effect of LID on the IV curve is shown in Figure 7.3c.

In contrast to the other degradation mechanisms, there is no direct analytical equation for the degradation rate due to LID ($k_{LID}(t)$). Instead, it can be derived from the change in power output as

$$k_{LID}(t) = \frac{1 - \frac{P(t+\Delta t)}{P(t)}}{\Delta t}, \tag{7.9}$$

where $P(t)$ is the output power at time t, excluding the other losses.

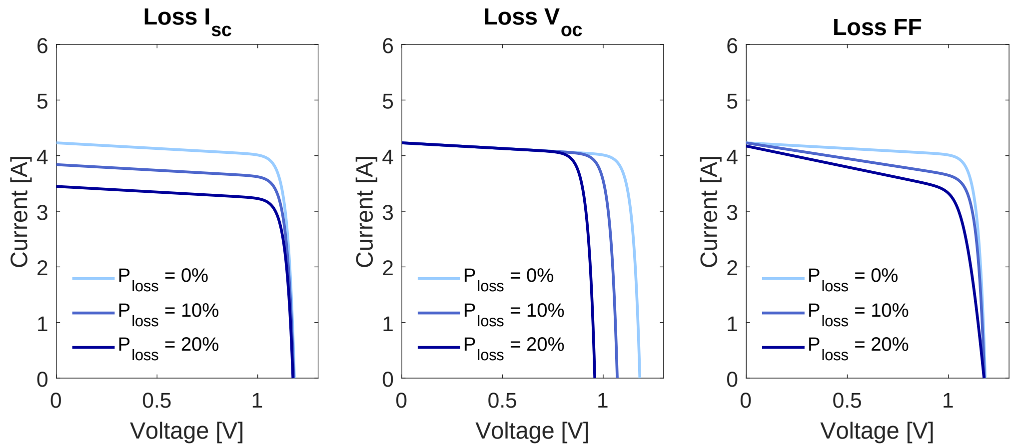


Figure 7.4: The effect on the IV curve of the perovskite subcell due to different degradation scenarios.

7.0.2 Scenario modeling

Degradation in the top cell is simulated using a scenario-based approach. Specifically, we introduce performance loss in the perovskite cell by adjusting parameters in the electrical model, following methods similar to those used by Qian *et al.* [70] and Orooj *et al.* [71]. To examine the effect on different degradation trends, three scenarios are considered where there is a loss in short circuit current (I_{sc}), open circuit voltage (V_{oc}), and fill factor (FF). These scenarios are simulated by adjusting I_{ph} (for I_{sc} degradation), I_0 (for V_{oc} degradation), and both R_{sh} as well as R_s (for FF degradation).

It is important to realize that the degradation of a perovskite cell in practice is not a loss in a single component, but rather a combination of multiple losses due to different mechanisms. The reason for analyzing these extreme cases is to establish a range of tolerable degradation rates that encompasses all possible degradation scenarios. If a real-world scenario involves a combination of the tested cases, its tolerable degradation rate will fall somewhere between the values derived in this study.

Figure 7.4 illustrates the impact of these scenarios on the IV curve for various values of P_{loss}. To simulate lifetime degradation, different constant values for k_{per} are applied. To indicate a specific degradation scenario, the notation $k_{per,X}$, where X can be I_{sc}, V_{oc} or FF, is used. The corresponding P_{loss} for the top cell is calculated using Equation (7.1), enabling the necessary adjustments to the IV curve.

It should be noted that this scenario based approach only considers fixed degradation rates over the lifetime of the PV module. This means that non-linear trend in the degradation of the perovskite cell are not captured by this method.

8. Conversion

The final step in simulating the energy yield of a PV system involves modeling the conversion of direct current (DC) to alternating current (AC). The PVMD Toolbox supports various inverter topologies, including central, string, and micro-inverters, allowing for flexible system configurations. For each topology, the AC output power (P_{AC}) is calculated with the SNL model [72] This model expresses P_{AC} as a function of the DC input power, DC input voltage, and inverter-specific parameters, capturing the efficiency and operational characteristics of the inverter.

The required DC input power and voltage are obtained from the module IV curves computed in the previous simulation step. Depending on the selected inverter topology, the module voltages and currents are aggregated using the series and parallel connection rules defined in Equations (6.5) and (6.6), respectively.

9. Financial

10. Loss Analysis

In Loss Analysis model, 17 different loss mechanisms are defined such that the sum of all the losses and the AC electricity equal the incoming in-plane irradiance. The 17 losses are divided over 4 different categories, as shown in Figure 10.1. The losses are calculated per category from left to right, starting with the fundamental losses. In the rest of this section, the equations for every loss mechanism are defined and the limitations of the model are discussed.

The fundamental losses

We consider fundamental losses as all losses considered for the detailed balance limit [73]. The equations of these fundamental losses are mostly derived from the work of Hirst et al [74]. The considered fundamental losses are the thermalization, below bandgap, emission, Carnot, and angle mismatch losses. For the fundamental losses, the bandgap energies of perovskite (1.68 eV) and silicon (1.12 eV) are used. Furthermore, an ideal solar cell is considered, meaning it has a perfect external quantum efficiency (EQE) and no non-radiative recombination. A perfect EQE is defined as having a value of 1 up to the wavelength corresponding with the bandgap energy. The thermalization loss of a single photon is the difference between its energy and the bandgap energy of the solar cell [36], which will be dissipated as heat. The total thermalization losses (P_{term}) can then be calculated with an integral over all wavelengths, as shown in Equation (10.1). The integral should be taken from over all wavelengths smaller than the wavelength corresponding to the bandgap energy (λ_g), written as

$$P_{term} = A_{mod} \cdot \int_0^{\lambda_g} \phi_{in}(\lambda) \cdot \left(\frac{h \cdot c}{\lambda} - E_g \right) d\lambda, \tag{10.1}$$

where A_{mod} is the area of the module, ϕ_{in} is the photon flux for a given wavelength, h and c are Plank's constants and the speed of light respectively, and E_g is the bandgap energy.

The below bandgap losses are accounting for all photons that cannot be absorbed due to insufficient energy [36]. This loss can be calculated by integrating over all wavelengths larger than λ_g, written as

$$P_{below} = A_{mod} \cdot \int_{\lambda_g}^{\infty} I_{in}(\lambda) d\lambda, \tag{10.2}$$

where $I_{in}(\lambda)$ is the incoming irradiance for a given wavelength.

Like the sun, a solar cell also emits black body radiation. These emitted photons are lost and are considered as radiative recombination. The emission can be modelled with the generalized Plank equation [36, 74, 75]:

$$\phi_{em}(\lambda, T, \mu, \Omega) = \frac{2 \cdot \Omega \cdot c}{\lambda^4} \cdot \frac{1}{e^{\frac{E-\mu}{k \cdot T}} - 1} \tag{10.3}$$

where ϕ_{em} is the emitted photon flux, T is the temperature, μ is the chemical potential of the solar cell, and Ω is the solid angle. Because each emitted photon accounts for a loss equal to the bandgap energy, the emission losses can be written as

$$P_{emission} = A_{mod} \cdot \int_0^{\lambda_g} E_g \cdot \phi_{em}(\lambda, T_{cell}, q \cdot V_{opt}, \Omega) d\lambda, \tag{10.4}$$

where T_{cell} is the temperature of the cell calculated with the fluid dynamic model, q is the charge of an electron, V_{opt} is the optimal voltage that maximizes the power for an ideal solar cell [74]. Ω is the solid state of emission, which is 2π for mono-facial cells and 4π for bifacial cells [76–78].

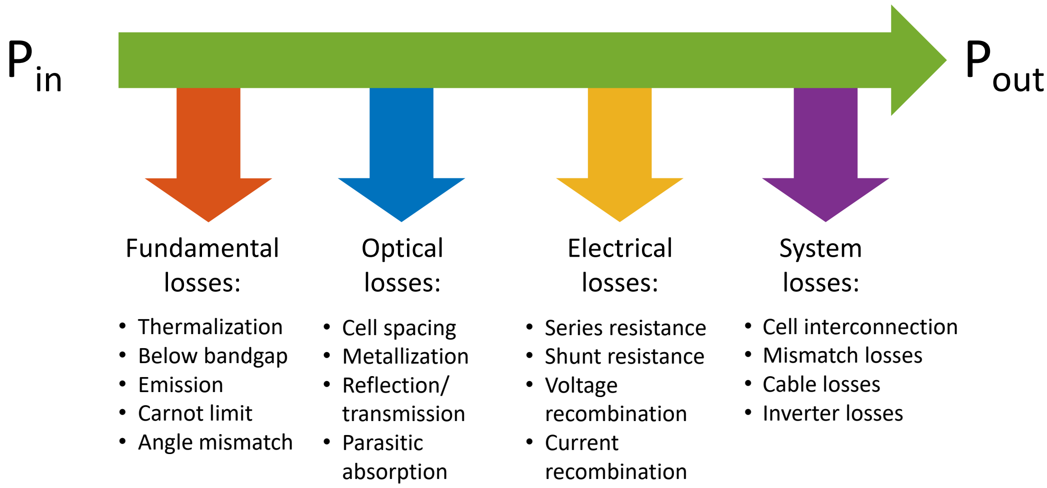


Figure 10.1: The 17 different losses that are considered. The losses are calculated from left to right.

The optimal voltage discussed before is lower than $\frac{E_g}{q}$. This voltage difference can be divided into the Carnot voltage loss (V_{Carnot}) and the angle mismatch voltage loss (V_{angle}). The Carnot loss (P_{Carnot}) and the angle mismatch loss (P_{angle}) can be calculated with

$$P_{carnot} = V_{Carnot} \cdot I_{max} \tag{10.5}$$

$$P_{angle} = V_{angle} \cdot I_{max}, \tag{10.6}$$

where I_{max} is the maximum output current and it is the difference between the incoming photon flux and the emitted photon flux. A quantification of the fundamental losses of an ideal solar cell under AM1.5g spectrum are shown in Figure 10.2. Note that the presented upper limits differ from the detailed balance limit [73] and the limit calculated by De Vos [79], since the AM1.5G spectrum is considered instead of the plank spectrum.

In addition to the five fundamental losses, a term called "non-ideality effect" is added to account for a non-ideal solar cell not matching the assumption used above. This includes that the emission losses are overestimated because the actual output voltage is typically smaller than the optimal voltage. It also accounts for absorption of photons with lower energy than the bandgap energy, as described by Urbach's rule [80, 81]. Lastly, it accounts for an underestimation of the thermalization losses, which is only included for tandem modules.

The optical losses

The optical losses account for all photons (with a higher energy than the bandgap energy) that are not absorbed by the absorber material. The optical losses include cell spacing, metal shading, reflection/ transmission, and parasitic absorption losses.

Both the cell spacing losses ($P_{cell-spacing}$) and metal shading losses (P_{metal}) can be considered as non-active area losses. The cell spacing losses occur, since the areas of the solar cell do not completely fill the module area, and it can be calculated as

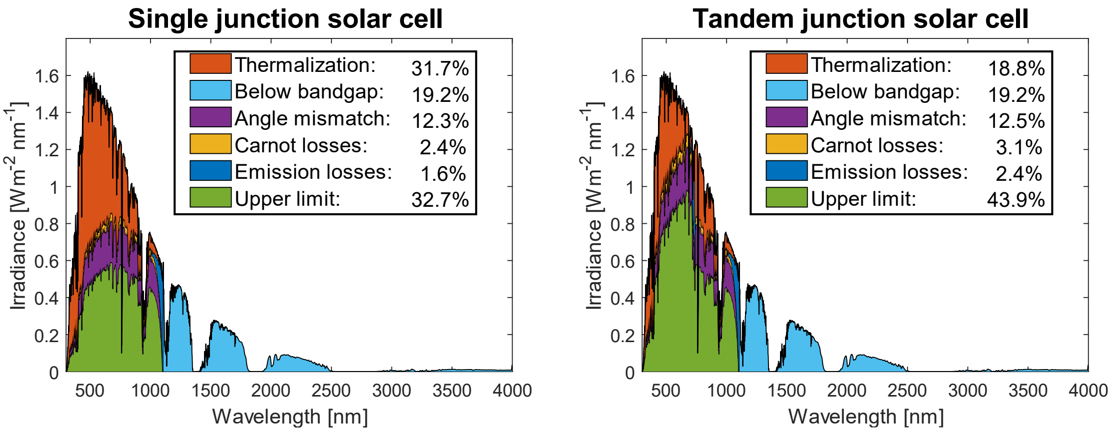


Figure 10.2: The fundamental losses and upper limit of a single junction Si (1.12 eV) solar cell and double junction perovskite (1.68 eV)/Si (1.12 eV) solar cell.

$$P_{cell-spacing} = (P_{in} - P_{fund}) \cdot \left(1 - \frac{N_{cells} \cdot A_{cell}}{A_{mod}}\right), \tag{10.7}$$

where P_{in} is the total incoming power, P_{fund} is the sum of the fundamental losses, N_{cells} is the number of cells, and A_{cell} is the area of a single cell.

The metal shading losses are calculated in a similar way as

$$P_{metal} = (P_{in} - P_{fund}) \cdot \frac{N_{cells} \cdot A_{cell}}{A_{mod}} \cdot SF, \tag{10.8}$$

where SF is the shading factor that indicates what fraction of the solar cells is effectively covered by metal contacts.

The losses due to reflection and transmission ($P_{ref,trans}$) are calculated based on the results of the optical model of the Toolbox, since this provides the reflectance and transmittance of every wavelength. It can be calculated with

$$P_{ref,trans} = A_{eff} \cdot \int_0^{\lambda_g} (\phi_{in}(\lambda) - \phi_{em}(\lambda)) \cdot (R(\lambda + T(\lambda)) \cdot q \cdot V_{opt} d\lambda, \tag{10.9}$$

where A_{eff} is the effective area of the module, defined as the total cell area minus the non-active area. $R(\lambda)$ and $T(\lambda)$ are respectively the reflection and transmittance for a certain wavelength. The term $q \cdot V_{opt}$ is used instead of the bandgap energy, because the Carnot and angle mismatch voltage are already subtracted from these photons.

The parasitic absorption losses ($P_{par,abs}$) account for absorption by all materials in the module except for the cell absorbers, e.g. silicon and perovskite. The reflectance and transmittance in Equation (10.9) are replaced with the parasitic absorptance (α_{par}), written as:

$$P_{ref,trans} = A_{eff} \cdot \int_0^{\lambda_g} (\phi_{in}(\lambda) - \phi_{em}(\lambda)) \cdot \alpha_{par} \cdot q \cdot V_{opt} d\lambda, \tag{10.10}$$

where the parasitic absorptance can be written as

$$\alpha_{par}(\lambda) = 1 - R(\lambda) - T(\lambda) - A_{si}(\lambda) - A_{pvk}(\lambda) \tag{10.11}$$

where $A_{si}(\lambda)$ and $A_{pvk}(\lambda)$ is the absorption in the silicon and perovskite layer respectively.

The electrical losses

After the optical losses, the electrical losses are calculated, which consist of resistive losses and recombination losses. The electrical characterization of the top and bottom cell is done using our advanced semiconductor analysis (ASA) software [82]. IV curves of top and bottom cell are simulated for various operating conditions (i.e. different temperatures (-15 °C - 95 °C) and different irradiances(50-1500 $\mathrm{W\,m^{-2}}$) such that all real-world conditions are included) by solving the Poisson equation and continuity equations in one dimension. To accelerate the computation, a one-diode equivalent circuit model [36–39] of the solar cell is fitted to match all the IV curves accurately [7]. This provides a Calibrated Lumped-Element Model (CLEM) with temperature and illumination dependent parameters [7]. The parameters included in this model are the photo-generated current (I_{ph}), the diode ideality factor (n), saturation current (I_0), series resistance (R_s), and shunt resistance (R_{sh}). Top and bottom cell IV-curves are combined to obtain the IV curve of the tandem cell by creating a series connection. The electrical losses are calculated based on the temperature and irradiance dependent values of the five parameters.

The series resistance losses (P_{series}) and the shunt resistance losses (P_{shunt}) are respectively calculated with the following equations:

$$P_{series} = I_{mpp}^2 \cdot R_s \tag{10.12}$$

$$P_{shunt} = \frac{(V_{mpp} + I_{mpp} \cdot R_s)^2}{R_{sh}} \tag{10.13}$$

In these equations, V_{mpp} is the maximum power point voltage, I_{mpp} is the maximum power point current.

Besides radiative recombination, it is possible for electrons and holes to recombine without emitting a photon. This is known as non-radiative recombination [36, 83]. This non-radiative recombination results in a voltage loss (V_{NRRV}) and a current loss (I_{NRRV}). The voltage loss is the difference between the optimal voltage and the voltage over the current source in the five-parameter model. The non-radiative recombination voltage loss (P_{NRRV}) can be written as

$$P_{NRRV} = (V_{opt} - V_{mpp} - V_{series}) \cdot I_{ph}, \tag{10.14}$$

where V_{series} is the voltage over the series resistance. The current loss can be modelled as the current flowing into the diode. The non-radiative recombination current loss (P_{NRRI}) can be calculated as

$$P_{NRRI} = (I_{ph} - I_{mpp} - I_{shunt}) \cdot (V_{mpp} + V_{series}) \tag{10.15}$$

where I_{shunt} is the current flowing into the shunt resistance.

The system losses

The last category is the system losses. The considered losses are the interconnection, mismatch, cable, and inverter losses. The interconnection losses (P_{int}) occur since the interconnection between the cells typically has some resistance. The ohmic losses of the interconnection can be written as

$$P_{int} = I_{mod}^2 \cdot R_{int}, \tag{10.16}$$

where I_{mod} is the output current of the module, and R_{int} is the resistance of the interconnection.

Mismatch losses are caused by differences in the maximum power point conditions among the cells. Because the cells are coupled, the operating conditions of an individual cell might differ from its maximum power point conditions. The mismatch losses (P_{mism}) are calculated as the difference between the sum of the individual maximum power points and the actual output power:

$$P_{mism} = \sum_{i=1}^{N_{cells}} (P_{mpp,i}) - I_{mod} \cdot (V_{mod} + I_{mod} \cdot R_{int}) \,. \tag{10.17}$$

In this equation, P_{mpp} is the maximum power point power of an individual cell i, and V_{mod} is the voltage of the module. The voltage over the interconnection has to be added to the module voltage, since this is the voltage over all the cells. Mismatch losses are especially important for two-terminal tandem modules, since the top and bottom cell need to be current matched.

When the DC electricity is converted into AC electricity, cables are installed, which typically introduce losses. The cable losses (P_{cable}) are calculated as

$$P_{cable} = P_{in,cable} \cdot (1 - \eta_{cable}) \,, \tag{10.18}$$

where $P_{in,cable}$ is the electricity flowing into the cables, and η_{cable} is the efficiency of the cables that is provided by the inverter model of the Toolbox.

The last considered loss is the inverter loss. The inverter losses ($P_{inverter}$) are calculated by taking the differences between the incoming electricity ($P_{DC,in}$) and the final AC electricity (P_{AC}), written as

$$P_{inverter} = P_{DC,in} - P_{AC} \,. \tag{10.19}$$

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